

Information, State and Process

FFGFT's Position on the Physical Reality of Information
Landauer, Magnetism, the Casimir Effect, and $\tilde{T} \cdot m = 1$

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Stored bit: “carries no additional energy or mass”, “does not weigh more than an empty one” → carries only the tiny electrostatic state energy, does not weigh measurably more ($\sim 10^{-16}$ J or $\sim 10^{-33}$ kg per Flash cell).
- Casimir effect (1 Oct. 2026): “the geometry stores no energy” → the configuration has the energy $E/A = -\pi^2 \hbar c / (720 z^3)$, but consumes none while the plates are at rest.

Abstract

Is information energy? The question is often answered affirmatively in information physics. FFGFT differentiates fundamentally:

- The mere *storage* of information as a state is energy-neutral.
- Only *processing* — in particular *erasure* — carries an energy dimension.

This distinction is not philosophical but empirically decidable: permanent storage media, magnetic order, and the Casimir effect confirm it directly. In FFGFT, states are geometric configurations on T^4 ; energy is a derived quantity that follows from processes, not from the bare existence of structure.

.1 The Question

In information physics — represented by J. A. Wheeler’s “It from Bit” [6] and in the IPI context by Denis [9] — information is often treated as the ontological primitive from which energy and matter follow. Landauer’s principle [1] is frequently cited as evidence for the physical reality of information: since erasing a bit costs energy, information must be physical.

This argument conflates two structurally distinct concepts:

Central Distinction

Information as state — a configuration, a stored bit, a winding number — exists without energy expenditure. It is a topological or geometric property.

Information as process — reading, writing, erasing, transforming — costs energy. Landauer’s principle applies *only* to erasure, i.e. to a *process*, not to the state itself.

Anyone who conflates these must accept a direct empirical consequence: every stored bit on a switched-off device would continuously consume energy — which would make permanent storage media fundamentally impossible. The existence of such media refutes the equation of information and energy.

.2 Landauer’s Principle — Precisely

Rolf Landauer showed in 1961 that the irreversible erasure of one bit dissipates heat into the surroundings [1]. The minimum energy required is:

$$E_{\min} = k_B T \ln 2 \quad (1)$$

where k_B is Boltzmann's constant and T is the absolute temperature. At room temperature this corresponds to approximately 2.9×10^{-21} J per erased bit.

What Landauer shows: Erasure — an irreversible process — dissipates energy. This is a consequence of the second law of thermodynamics [2].

What Landauer does not show: That the bare *existence* of a stored bit requires energy. A magnetic bit on a hard drive, a winding number on T^4 , a spin alignment in a permanent magnet — all exist without continuous energy input. Only the transition between states costs energy.

In natural units ($\hbar = c = k_B = 1$), the erasure energy has the dimension of a temperature, hence a mass, hence an energy — connecting to $\tilde{T} \cdot m = 1$ in FFGFT. But this dimensional relationship holds for the *process*, not for the resting state.

.3 Writing versus Erasing: No Symmetric Energy Requirement

One might object: writing a bit also costs energy — and if energy is introduced, mass arises via $E = mc^2$. Would information then carry mass after all?

This conclusion is wrong for two reasons.

Logical Reversibility Decides, Not the Process as Such

The four possible deterministic operations on one bit are: HOLD, RESET-to-0, RESET-to-1, and NOT (flip). RESET-to-0 and RESET-to-1 are logically *irreversible* — the previous state is lost. NOT and HOLD are logically *reversible* — the previous state can be reconstructed from the result.

Landauer's bound applies *only* to logically irreversible operations, i.e. only to RESET [1]. Dago and Bellon demonstrated experimentally that the NOT operation — writing a new state by bit-flip — can be carried out in a physically reversible procedure at vanishingly small energy cost, far below the Landauer bound [11]. There is no fundamental energy bound for writing as such.

Energy Flows into the Environment, Not into the State

In practice, writing always costs energy — but this energy does not go *into the stored bit*. It goes into:

- Ohmic heat in the write current (Flash, DRAM)
- Magnetic remagnetisation work against the coercive force
- Mechanical dissipation in the write head

The stored bit itself carries only the tiny electrostatic energy of its state, not the energy spent on writing. A written Flash memory does not weigh measurably more than an empty one — even though writing cost energy. (*Corrected on 30 Sept. 2026: previously "carries no additional energy or mass" and "does not weigh more than an empty one" – a charged floating gate stores roughly $\frac{1}{2}QV \approx \frac{1}{2} \cdot 1.6 \times 10^{-16} \text{ C} \cdot 5 \text{ V} = 4 \times 10^{-16} \text{ J}$, i.e. $\Delta m = E/c^2 \approx 4 \times 10^{-33} \text{ kg}$ per cell; tiny, but not zero.*) The energy has flowed into the environment, not been stored in the state.

Why Information Does Not Create Mass

The inference "writing costs energy $\rightarrow E = mc^2 \rightarrow$ mass arises" commits a category error. $E = mc^2$ applies to the rest energy of a physical system — i.e. to the mass of the storage

medium itself, not to the energy that dissipates into the environment during the write process.

In FFGFT this is consistent: winding numbers on T^4 are topological invariants. The transition from one configuration to another is a process with an energy dimension via $\tilde{T} \cdot m = 1$ — but the configuration itself carries no additional rest mass. The process does not create new mass; it changes the state of a system that already has mass.

.4 Three Empirical Pillars for the Energy-Neutrality of States

Pillar 1: Magnetism

A permanent magnet is the most tangible evidence for the energy-neutrality of information states. Spins in a ferromagnetic material are aligned — an information state in the physical sense (north vs. south pole, bit = direction of magnetisation).

- **Storage:** requires no ongoing energy. The magnet holds its alignment without power supply — in principle indefinitely.
- **Remagnetisation:** costs energy. Overcoming the coercive field strength requires an external field that performs work.
- **Slow decay:** Over very long timescales, thermal fluctuations, cosmic radiation, or other external disturbances can alter the alignment [7]. But this decay is externally triggered — a process with an external cause, not an intrinsic property of the state itself.

This is the structural point: even the slow decay of a permanent storage medium is an externally triggered process, not evidence that the state itself costs energy.

Pillar 2: The Casimir Effect

The Casimir effect shows that vacuum geometry has physical consequences without any classical energy store being present. Two uncharged, parallel metal plates in vacuum attract one another — a measurable force first predicted in 1948 by Hendrik Casimir [4] and first precisely measured in 1996 by Steve Lamoreaux [5].

The force per unit area is:

$$P(z) = -\frac{\pi^2}{240} \frac{\hbar c}{z^4} \quad (2)$$

where z is the plate separation.

What the Casimir effect shows: The geometric restriction of the allowed vacuum modes between the plates produces an energy density difference that generates a force. The *geometry of the vacuum* — the state — has physical consequences.

What the Casimir effect does not show: That this geometry consumes energy while the plates are stationary. The configuration has an energy, $E/A = -\frac{\pi^2}{720} \frac{\hbar c}{z^3}$, whose derivative $-\partial_z(E/A)$ is the force per unit area above; this energy stays constant at fixed z . Only the *motion of the plates* — the process — performs or absorbs work. (Corrected on 1 Oct. 2026: previously “That this geometry stores or consumes energy” – the plate configuration has the Casimir energy $E/A = -\pi^2 \hbar c / (720 z^3)$, and $-\partial_z(E/A) = -\pi^2 \hbar c / (240 z^4)$ is exactly $P(z)$; the geometry therefore has an energy, it merely does not consume any.)

The structural parallel to FFGFT is this: the winding structure on T^4 is a geometric configuration — it exists, has consequences (mass ratios, coupling constants), but it does not cost ongoing energy. Only dynamic transitions between configurations — processes — carry an energy dimension. This parallel is an analogy of structure, not an identity of mechanism.

Pillar 3: Permanent Storage Media

The entire digital civilisation rests on the energy-neutrality of stored information. Flash memory, magnetic hard drives, optical data carriers — all use states that persist without power supply.

- **Flash memory** stores bits as charge states in floating-gate transistors. The charge is retained without energy input — typically for ten years or more.
- **Magnetic tapes** hold data for decades without power.
- **Slow decay:** Quantum tunnelling, thermal activation, and ionising radiation can flip bits over very long timescales [8]. But here too: every individual flip is an externally triggered process, not a spontaneous property of the resting state.

If information as a state were not energy-neutral, no permanent storage medium would be possible. The existence of this technology is the empirical proof of the central distinction.

.5 In Natural Units: State, Process, and the Foundational Relation

In natural units ($\hbar = c = k_B = 1$), dimensions collapse:

$$[E] = [m] = [T^{-1}] = [\text{temperature}] \quad (3)$$

The erasure energy $E_{\min} = k_B T \ln 2$ becomes in natural units $E_{\min} = T \ln 2$ — a temperature, a mass, an energy.

FFGFT's foundational relation $\tilde{T} \cdot m = 1$ connects time and mass directly. A state transition — a process — has a duration and changes the effective mass of the dynamical field $\Delta m(x, t)$. This is the natural embedding of Landauer's principle in FFGFT:

- **State:** Winding configuration on T^4 — m is geometrically fixed by ξ , no energy consumption.
- **Process:** Transition between configurations — Δm changes; via $\tilde{T} \cdot m = 1$ this has a time dimension and hence an energy dimension.

The energy cost of information processing is therefore not an external addition in FFGFT — it follows from the foundational relation.

.6 FFGFT's Position

FFGFT and Information

Primitive: Geometry — T^4 winding structure, ξ , $\tilde{T} \cdot m = 1$.

State: Winding configuration — no energy primitive, analogous to the magnetic bit or Casimir geometry.

Process: State transition — carries an energy dimension, follows from $\tilde{T} \cdot m = 1$ and ξ .

Information: A descriptive layer over geometric reality, not an ontological primitive.

Shannon information [3] — bits, entropy $H = -\sum p_j \log p_j$ — is a counting quantity, dimensionless. It describes states in a probability space. It is not a primitive in FFGFT because FFGFT does not use probabilities as a foundational concept — winding numbers are deterministic topological invariants, not random variables.

.7 Demarcation

Wheeler — “It from Bit”

John Archibald Wheeler’s position [6]: information is the ontological primitive from which physical reality arises. This places information before geometry.

FFGFT inverts this: geometry (T^4 , ξ) is the primitive. Information in the Shannon sense is a description of configurations in this geometry, not a cause of it.

Denis — Landauer Vacuum (IPI)

Denis [9] proposes deriving the cosmological constant via Landauer’s principle from an information-theoretic vacuum sector. This presupposes that information is ontologically fundamental.

FFGFT derives $\Lambda = 1.34 \times 10^{-122}$ (Planck units) directly from ξ [12], without information theory as an intermediate layer. The path is shorter and requires no Landauer bridge.

RSG — Survival Weights

RSG [10] uses survival weights $S_i = e^{-A_i}$ for histories. This is a probability structure over configurations — a descriptive layer, not a geometric primitive. From FFGFT’s perspective, RSG is a possible description of selection processes, but not an explanation of the scale values that determine Γ_i .

.8 Symbol List

Symbol	Meaning
k_B	Boltzmann constant: $k_B = 1.380649 \times 10^{-23}$ J/K
T	Absolute temperature (Kelvin)
$E_{\min} = k_B T \ln 2$	Landauer bound: minimum energy to erase one bit
$H = -\sum p_j \log p_j$	Shannon entropy; dimensionless in bits (base 2) [3]
$\xi = 4/30000$	FFGFT coupling parameter; dimensionless
$\tilde{T} \cdot m = 1$	FFGFT foundational relation (natural units)
T^4	4-dimensional torus; FFGFT ontological primitive
$\Delta m(x, t)$	Dynamical mass field in T0; carries state transitions
$P(z) = -\frac{\pi^2}{240} \frac{\hbar c}{z^4}$	Casimir pressure; z = plate separation [4]
$\Lambda = 1.34 \times 10^{-122}$	Cosmological constant from ξ (Planck units) [12]

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